

ARK-459 — Preregistration

Cloud IAM Role Grant · Revocation At Execution

Series: ExecutionProof authorization-boundary corpus — **Production-Boundary phase (P01)**

Experiment ID: ARK-459

Institution: Remnant Fieldworks Inc.

PI: Derek Adam Hone

Substrate: Classical software (no quantum hardware, no cryptographic security claims)

Guards: Dual independent — V1 (JavaScript), V2 (Python)

Preregistration status: LOCKED before official execution (see `MANIFEST.txt`)

Commercial / standards / IP justification (required by RF covenant)

- **Commercial purpose:** Directly answers **Prospect Question #4** — “What happens when authority is revoked after approval but before execution?” — in the **recognizable AWS-style IAM context** established by ARK-458. Buyers ask this constantly: the authorization path splits approval ($t=0$) from execution ($t=\text{later}$), and revocation can land in that window. A correct gate must **fail closed** (DENY when revocation is effective) or **fail safe** (HOLD when in-flight), never silently ALLOW as if the authority were still valid.
 - **RF-100 requirement:** Exercises the **re-check-at-execution** conformance requirement — that an approval does not grant perpetual authority; the gate must re-verify validity at the moment of resource contact.
 - **Patent-family connection:** Produces dated **working-example** evidence for the authority-revocation / fail-closed / HOLD mechanisms described in the filed parent application 19/529,283 and the AI-governance / proof CIP family. Working-example evidence only — it does not legally validate any claim, and adds no new matter to any filed application.
-

Question

When an IAM role-grant authorization is **APPROVED** (at `t_approval`), but the approving authority is **REVOKED** before the grant is executed (at `t_execution`), does an independent execution gate:

- **DENY** when the revocation is **fully effective** before `t_execution` (fail-closed)?
- **HOLD** when the revocation is **in-flight** (issued but not yet fully propagated) at `t_execution` (fail-safe — cannot confirm validity)?
- **ALLOW** only when authority is valid at execution (no revocation, revoked only after execution, or a valid reauthorization issued after the revocation)?

This extends ARK-451 (authority revocation) to the **production IAM boundary** (ARK-458’s 5-dim action tuple). The IAM action itself is exact-match on all five dimensions; only the **timeline and revocation** vary across arms.

Model

Each scenario describes one IAM role-grant authorization lifecycle:

IAM Action Tuple (from ARK-458, exact-match)

Dimension	Role
<code>principal</code>	ARN of the grantee (who receives the role)
<code>role</code>	the role / managed policy being granted
<code>account</code>	target AWS account id
<code>permission_set</code>	canonical hash of the effective permission scope
<code>condition</code>	scope condition (region / MFA / resource constraint)

The execution action matches the approved action **exactly** on all five dimensions. This is NOT an exact-action-binding test (ARK-458 covered that); this is a **revocation-timing** test.

Timeline

- `t_approval` — the IAM grant is approved
- `t_execution` — the grant is attempted (the moment the action contacts the IAM service)
- `revocation` — `null`, or `{ t_revoke, propagation_delay, reason }`
- `reauthorization` — `null`, or `{ t_reauth, valid }`
- `multistep` — whether the grant is the irreversible final step of a multi-step workflow

A revocation becomes **fully effective** at `eff = t_revoke + propagation_delay`. The `propagation_delay` models the real-world gap between when a revocation is issued (e.g., credential termination entered into the identity provider) and when it is observable at the IAM enforcement point.

Decision procedure (both guards, identical)

Evaluated at `t_execution` (re-check at moment of IAM grant attempt):

1. If `revocation` is `null` → **ALLOW** (authority valid throughout).
 2. Compute `eff = t_revoke + propagation_delay`.
 3. If a `reauthorization` exists, is `valid`, and `t_revoke < t_reauth ≤ t_execution` → **ALLOW** (a new, independent approval governs).
 4. Else if `eff ≤ t_execution` → **DENY** (authority provably revoked before execution; fail closed).
 5. Else if `t_revoke ≤ t_execution < eff` → **HOLD** (revocation in-flight / unconfirmed at execution; fail safe).
 6. Else (`t_revoke > t_execution`) → **ALLOW** (revocation issued only after the grant already executed under valid authority).
-

Arms

8 arms × 100 scenarios per arm = **800 evaluation decisions**.

Arm	Label	Timeline constraint	Expected
1	VALID-throughout	no revocation	ALLOW
2	REVOKED-before-ap-proval	$\text{eff} < \text{t_approval}$	DENY
3	REVOKED-after-ap-proval-before-execution	$\text{t_approval} \leq \text{t_revoke}$, $\text{eff} \leq \text{t_execution}$	DENY
4	REVOKED-during-multistep	$\text{multistep} = \text{true}$, $\text{eff} \leq \text{t_execution}$	DENY
5	IN-FLIGHT-at-execution	$\text{t_revoke} \leq \text{t_execution} < \text{eff}$	HOLD
6	REVOKED-then-REAUTHORIZED	valid reauth with $\text{t_revoke} < \text{t_reauth} \leq \text{t_execution}$	ALLOW
7	IN-FLIGHT-boundary	$\text{t_revoke} \leq \text{t_execution} < \text{eff}$ (boundary case, longer propagation)	HOLD
8	REVOKED-after-execution	$\text{t_revoke} > \text{t_execution}$	ALLOW

Randomized per scenario: all IAM action dimensions, all timeline offsets, `propagation_delay`, revocation `reason` — subject to the arm's timing constraint.

Revocation-timing gate (anti-no-op oracle)

An **independent structural oracle** (`_revocation_effective_oracle`) verifies, for every generated scenario, that it genuinely encodes its arm's timing relationship before any guard sees it:

- Arm 1: `revocation is null`.
- Arm 2: $\text{eff} < \text{t_approval}$, no reauth.
- Arm 3: $\text{t_approval} \leq \text{t_revoke}$ and $\text{eff} \leq \text{t_execution}$, no reauth.
- Arm 4: $\text{multistep} == \text{true}$ and $\text{eff} \leq \text{t_execution}$, no reauth.
- Arms 5, 7: $\text{t_revoke} \leq \text{t_execution} < \text{eff}$ (genuinely in-flight), no reauth.
- Arm 6: valid reauth with $\text{t_revoke} < \text{t_reauth} \leq \text{t_execution}$.
- Arm 8: $\text{t_revoke} > \text{t_execution}$, no reauth.

If **any** scenario fails its check, the generator raises `ValueError` (the run aborts). This prevents ARK-455-style defects where a case appears to encode a condition but is mathematically inert.

Metrics

- **g_allow_valid_min** = minimum ALLOW rate across arms {1, 6, 8} (expect ALLOW when valid at execution)
- **g_deny_revoked_min** = minimum DENY rate across arms {2, 3, 4} (expect DENY when revoked before execution)
- **g_hold_inflight_min** = minimum HOLD rate across arms {5, 7} (expect HOLD when in-flight)
- **g_margin** = $\min(\text{g_allow_valid_min}, \text{g_deny_revoked_min}, \text{g_hold_inflight_min}) - 0.05$
- Secondary: V1 $\leftrightarrow$ V2 decision concordance (target 100%).

Pass criteria (fixed before execution)

- **C1:** `g_allow_valid_min` ≥ 0.95
- **C2:** `g_deny_revoked_min` ≥ 0.95
- **C3:** `g_hold_inflight_min` ≥ 0.95
- **C4:** `g_margin` ≥ 0.90
- **Verdict** = **PASS** iff $C1 \wedge C2 \wedge C3 \wedge C4$, else FAIL. Result is published either way.

Kill conditions / effectiveness gate

1. **Effectiveness oracle:** every scenario must genuinely encode its arm's timing relationship (verified by `_revocation_effective_oracle` before any guard sees it). Any violation $\Rightarrow$ abort.
2. **Kill-gate (negative control):** a deliberately broken guard that **ignores revocation** and always **ALLOWS** must wrongly ALLOW $> 50\%$ of the DENY/HOLD arms {2,3,4,5,7} — proving the testbed is falsifiable. If the broken guard does not fail, the test is void.

Publication rule

The outcome (PASS, FAIL, or ABORT) will be published to GitHub with raw per-arm results, and defensively published to Zenodo (CC BY 4.0), regardless of verdict.

Honest bounds

- Classical software test of **authorization control logic**, not a test of AWS, not a cryptographic security proof, not a production-readiness certification.
 - This tests the **timeline/revocation logic** in the IAM context; ARK-458 already tested exact-action-binding.
 - Results are bounded to this scenario model, these arms, and this seed family.
 - No claim that this experiment legally validates any patent claim or certifies RF-100 conformance.
-

Preregistered under the Remnant Fieldworks Standing Covenant (preregister $\rightarrow$ lock $\rightarrow$ execute $\rightarrow$ publish all outcomes). To God be the glory. Proof Before Power. Verification Before Execution.